

TXA: timing and regimen

Adults with active/suspected major traumatic bleeding; time from injury.

≤3 h; 1 g IV over ≥10 min → 1 g IV over 8 h

Give early; Chinese R33 states at least eight hours for maintenance, European R23 eight hours.

After three hours, possible fibrinolytic shutdown or special populations:
source/local-policy review.

Follow local product, preparation and infusion policy; avoid rapid IV push.

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

European major bleeding guideline, sixth edition (2023)

Adult bleeding: BP target

Early adult bleeding before control, without brain/spinal injury.

SBP 80–90 mmHg

Caution with older age/chronic hypertension; separate pediatric/pregnancy protocols.

Worsening perfusion or neurological injury: revise targets and escalate bleeding control.

Chinese R22 / European R13 support conditional use, not a universal target.

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

European major bleeding guideline, sixth edition (2023)

Shock index

Adult risk recognition, with examination and trends.

SI = HR / SBP; Chinese SI ≥ 1

Chinese R2 includes ≥ 1 in diagnostic criteria; European R5 discusses risk around 0.9–1.

Blunted HR or apparently normal older-adult vitals do not exclude shock.

Do not use SI alone to exclude bleeding or decide transfusion.

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

European major bleeding guideline, sixth edition (2023)

MTP activation assessment

Adults; ACS TQIP example.

ABC ≥ 2 + clinical state

One point each: penetrating torso, positive FAST, HR >120 , SBP <90 ; verify local version.

Persistent instability/active bleeding may also trigger assessment; do not wait for cumulative volume.

Chinese guidance supports early blood/local policy; ABC is not claimed as a national activation rule.

ACS TQIP massive transfusion guidance

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

NHC clinical blood use technical standard (2025)

Fibrinogen

Major active bleeding or viscoelastic evidence of functional deficit.

Clauss Fib ≤ 1.5 g/L + major bleeding

European R29: assess fibrinogen/cryo replacement and retest.

Ongoing bleeding/worsening tests: blood-bank/coagulation escalation.

Chinese perioperative consensus uses <1.5 for cryo in severe oozing and <1.5 – 2.0 for fibrinogen; scope/operators differ.

European major bleeding guideline, sixth edition (2023)

Chinese anesthesia consensus on perioperative bleeding

Hb target during transfusion

European R16 target for adult trauma requiring RBC transfusion.

Hb 70–90 g/L

Use bleeding/perfusion/trajectory; normal early Hb does not exclude hemorrhage.

TBI, myocardial ischemia and special populations need specific goals; do not await Hb fall.

Chinese 2025 blood-use standard requires individual assessment/response review, not a national 70–90 target.

European major bleeding guideline, sixth edition (2023)

NHC clinical blood use technical standard (2025)

Ionized calcium

Adult major bleeding, especially massive transfusion.

iCa: maintain the local normal range

Recheck blood-gas iCa with pH; correct under local replacement policy.

Ongoing transfusion, falling iCa or rhythm changes: prompt reassessment.

European R31 and Chinese perioperative 3.8 support monitoring/normalization; products/doses are local.

European major bleeding guideline, sixth edition (2023)

Chinese anesthesia consensus on perioperative bleeding

Severe TBI: perfusion target

Severe TBI (GCS ≤ 8); avoid permissive hypotension.

MAP ≥ 80 mmHg

Chinese 2023 R22 also gives SBP >110 ; 2026 prehospital R6 gives SBP ≥ 110 or MAP ≥ 80 .

Changing consciousness/pupils or unmet goals: address oxygenation/perfusion and escalate neurological care.

With ICP monitoring, individualize CPP and other goals; MAP is not a complete neurocritical protocol.

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Chinese expert consensus on prehospital major trauma (2026)

European major bleeding guideline, sixth edition (2023)

Older adults: occult shock warning

Older-trauma high-risk triage signal, not a stand-alone shock diagnosis.

Age ≥ 65 ; SBP < 110 mmHg

Ask about anticoagulants, baseline function, BP/HR drugs and head injury.

Low-energy injury with symptoms, anticoagulation or poor perfusion: early senior review.

The 110 signal comes from prehospital consensus; hip guidance is a domestic orthopedic pathway entry.

Chinese expert consensus on prehospital major trauma (2026)

NHC geriatric hip fracture care and management guideline (2022)

Anticoagulants: reversal pathway

Life-threatening/uncontrolled bleeding or urgent high-risk surgery; authorized team decisions.

Drug → last dose → bleeding severity

Warfarin: PCC + IV vitamin K; dabigatran: idarucizumab; Xa agents: specific reversal or PCC pathway.

Check renal function, INR/TT/anti-Xa and formulary; normal PT does not exclude DOAC effect.

Verify Xa agent, indication and availability locally; reversal is not automatic for every treated patient.

European major bleeding guideline, sixth edition (2023)

Chinese anesthesia consensus on perioperative bleeding

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Pregnancy trauma: maternal/obstetric care

Uterus at/above umbilicus; unsensitized Rh-negative abdominal/major trauma.

Mother first; left uterine displacement; Rh D

Manual displacement with effective CPR; obstetric anti-D assessment by gestation, sensitization, FMH and local policy.

Instability, pain/bleeding or fetal concern: trauma/obstetric escalation.

No equivalent Chinese guidance for these specific items was verified; international reference, local obstetric confirmation.

Queensland: traumatic injury in pregnancy

AHA 2025: cardiac arrest in pregnancy

Pregnancy arrest: resuscitative delivery

Fundus at/above umbilicus with no ROSC despite standard resuscitation.

Prepare at arrest recognition; goal delivery by ≤ 5 min

AHA 2025: organize obstetric/anesthetic/neonatal teams and resuscitative delivery at the arrest site.

Historical four-minute start/five-minute birth framing does not mean waiting four minutes to call or prepare.

Pre-plan teams, equipment and drills locally; not an automatic operation at every gestation.

AHA 2025: cardiac arrest in pregnancy

Children: weight and first aliquot

RCH major pediatric trauma with poor circulation; assess bleeding in parallel.

Measured/recent weight; crystalloid 10 mL/kg

If unweighable use a locally validated length-based tool; RCH ongoing bleeding: initial RBC aliquot 10 mL/kg.

Reassess each aliquot; ongoing hemorrhage needs blood-bank/pediatric MTP, avoiding repeated large crystalloids.

Chinese major-hemorrhage component regimen is on the next card; routine RBC table excludes emergency major bleeding.

RCH: major pediatric trauma primary survey

WS/T 795—2022 pediatric transfusion guideline

Children: Chinese hemorrhagic transfusion index

WS/T795—2022 §9.5.2.2: pediatric traumatic hemorrhagic shock.

RBC : FFP = 1 : 1 (volume)

Standard example: RBC 20 mL/kg + FFP 20 mL/kg; not routine anemia/non-trauma ratio.

Persistent bleeding: additional components under standard/local MTP; adjust to labs and monitor overload.

A component-volume regimen, at a different stage from RCH initial aliquots; check product/local pediatric policy.

WS/T 795—2022 pediatric transfusion guideline

Burns: Chinese rule of nines

Head/face/neck; both arms; trunk including perineum; both legs including buttocks/feet.

Adult: $9 + 18 + 27 + 46 = 100\%$

Under 12: head/neck= $9+(12-\text{age})\%$; both legs= $46-(12-\text{age})\%$.

Uncertain area/depth, key-site/inhalation/electrical/chemical injury: burn review.

2025 standard Table 1 / §2.1; fluid-formula area uses second/third degree, excluding superficial burns.

NHC emergency burn treatment standard (2025 edition; published 2026)

Burns: Chinese adult fluid formula

Chinese 2025 crystalloid/colloid formula; second/third degree; enter percentage number.

First 24 h: $1.5-2 \text{ mL} \times \text{kg} \times \% \text{TBSA} + 2000 \text{ mL}$

Crystalloid:colloid=1-2:1; half estimated total in first eight hours from injury, quarters in next two intervals.

Titrate to perfusion, urine and overload; do not apply adult formula to children/cardiorenal disease.

1.5 is the lower end, not a fixed universal amount; include prior fluids and follow local burn policy.

NHC emergency burn treatment standard (2025 edition; published 2026)

Resuscitation, consent and handover records

Only when urgent rescue prevented timely writing; not permission to delay all records.

Truthful, marked supplementation ≤ 6 h after rescue ends

Minute-level time, actions, responses, staff, communication and handover ownership.

Record signing difficulties and use lawful emergency authorization; never fabricate signatures/times.

Record norms Article 22(8), 25/27; transfusion consent also follows 2025 blood-use standards.

Basic norms for medical record writing (2010)

NHC clinical blood use technical standard (2025)

Trauma bay: before arrival

Tiered recall; resident key-step simulation requires supervision and local policy.

Roles: head / chest / circulation / recorder

Confirm space, monitors, blood bank and senior support

Handover: mechanism, findings, actions, unknowns

Escalate immediate threats or inadequate resources

WHO trauma care checklist and team handover

Chinese expert consensus on prehospital major trauma (2026)

xABCDE: revisit each cycle

Tiered recall; resident key-step simulation requires supervision and local policy.

x bleeding → A airway/protection → B breathing → C circulation → D neurology → E exposure/warmth

Specific findings + trends + owner

Check effect after actions; revisit unknown regions

Deterioration: return to primary survey and escalate

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Pelvis: parallel risk recognition

Tiered recall; resident key-step simulation requires supervision and local policy.

Assess perfusion, open wounds and other bleeding sources

Binder skill: context, placement, review and failure escalation

Coordinate orthopedic / hemorrhage / urological / soft-tissue teams

Local cohort supports discussion, not individual decisions

Chinese expert consensus on prehospital major trauma (2026)

IJEM 2025 pelvic fracture polytrauma study

Care chain: continuity at transitions

Tiered recall; resident key-step simulation requires supervision and local policy.

Confirm destination capability and owner

Hand over knowns / unknowns / concerns / next review

Before movement check monitoring, airway, bleeding, equipment

Bottleneck or patient change: reassess destination

WHO trauma care checklist and team handover

Chinese expert consensus on prehospital major trauma (2026)

Disaster: patient and system review

Tiered recall; resident key-step simulation requires supervision and local policy.

Does demand exceed available capacity?

Update numbers, space, blood, surgery, transport

Patients: trends/findings; system: bottlenecks/hazards

Use institutional triage tools; reassess after change

WHO trauma care checklist and team handover

Resuscitation: parallel tasks

Tiered recall; resident key-step simulation requires supervision and local policy.

Hemorrhage control and resource coordination

Early TXA / blood / coagulation monitoring

Warmth, iCa, perfusion and response checks

See separate adult/TBI numerical cards

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

European major bleeding guideline, sixth edition (2023)

Secondary / tertiary: prevent missed injury

Tiered recall; resident key-step simulation requires supervision and local policy.

After stabilization: AMPLE + head-to-toe exam

Mark unassessed regions, pending imaging and limits

Repeat exam, reconcile final reports, assign ownership

New symptoms/deterioration: xABCDE review without waiting

RCH secondary survey

WHO trauma care checklist and team handover

Special populations: identify group

Tiered recall; resident key-step simulation requires supervision and local policy.

Older adults: anticoagulation, serious low-energy injury, baseline

Children: weight, age, blood protocol

Pregnancy: mother first, obstetric collaboration

Burns: depth, Chinese rule of nines, specialist review

- Chinese expert consensus on prehospital major trauma (2026)
- WS/T 795—2022 pediatric transfusion guideline
- NHC emergency burn treatment standard (2025 edition; published 2026)
- Queensland: traumatic injury in pregnancy

Handover: record and accept

Tiered recall; resident key-step simulation requires supervision and local policy.

Findings/time; actions/response

Unknowns and review deadlines

Communication / consent / receiver confirmation

See record card for six-hour rescue supplementation

Basic norms for medical record writing (2010)

NHC clinical blood use technical standard (2025)

Pelvic MDT case: staged debrief

Tiered recall; resident key-step simulation requires supervision and local policy.

Distinguish recorded facts from classroom simulation

Assess bleeding and open-wound risks in parallel

Explain collaboration points and reassessment ownership

Do not invent gaps; cohort membership unconfirmed

IJEM 2025 pelvic fracture polytrauma study

Skills: decision to reassessment

Tiered recall; resident key-step simulation requires supervision and local policy.

Timing → preparation/authorization → key steps

Effect checks → failure recognition → escalation

0 absent / 1 after prompting / 2 complete independently

Supervised simulation; scores do not authorize practice

NICE NG39 major trauma

TBI: neurology with ABC

Tiered recall; resident key-step simulation requires supervision and local policy.

Track consciousness, pupils and movement

Avoid hypoxia/hypoperfusion; see TBI pressure card

Check anticoagulation, drugs and baseline

New neurological change: senior airway/neurosurgical review

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Chinese expert consensus on prehospital major trauma (2026)

Chest: bilateral comparison

Tiered recall; resident key-step simulation requires supervision and local policy.

Work of breathing, chest excursion and breath sounds

Report bedside findings with monitor trends

Authorized operators; reassess after intervention

Rapid respiratory/perfusion decline: urgent team escalation

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Abdomen: consider occult bleeding

Tiered recall; resident key-step simulation requires supervision and local policy.

Mechanism, examination, perfusion and change

One negative FAST does not exclude abdominal injury

Repeat review/imaging by stability and local pathway

Deterioration: primary survey, bleeding and surgical review

NICE NG39 major trauma

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Polytrauma: global priorities

Tiered recall; resident key-step simulation requires supervision and local policy.

Keep axes visible; avoid fixation on one region

Assign owner and effect check to each action

After stabilization complete secondary/tertiary review

New findings/resources trigger reprioritization

WHO trauma care checklist and team handover

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Abdominal case: debrief trends

Tiered recall; resident key-step simulation requires supervision and local policy.

Separate known findings and assumptions

Connect perfusion, imaging and blood resources

State checks needed after each choice

No improvement: revisit bleeding sources and escalate

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

NICE NG39 major trauma

Shock reassessment: improvement?

Tiered recall; resident key-step simulation requires supervision and local policy.

Review perfusion, neurology, bleeding and warmth

Record action–time–response

Distinguish scripted trends from real treatment effects

Worsening/persistent unknowns: parallel team review

Chinese emergency consensus on traumatic hemorrhagic shock (2023)

Spine/hip: neurovascular recheck

Tiered recall; resident key-step simulation requires supervision and local policy.

Record pain, movement, sensation, distal perfusion

Document rechecks before/after movement or intervention

Link mechanism and examine other regions

New neurological/perfusion deficit: orthopedic/spine escalation

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Orthopedics: systemic then local

Tiered recall; resident key-step simulation requires supervision and local policy.

First identify immediate xABCDE threats

Local examination and neurovascular status

Recheck after temporary protection; look for associated injury

Open injury, ischemia or neurological deficit: senior review

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Spine: protection and serial neurology

Tiered recall; resident key-step simulation requires supervision and local policy.

Mechanism, pain, motor/sensory exam and reliability

Motion restriction by risk/local criteria

Record neurology before/after movement

New weakness/sensory change or poor perfusion: escalate

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Upper limb: pre/post comparison

Tiered recall; resident key-step simulation requires supervision and local policy.

Wound, deformity, movement, sensation, perfusion

Cover open injury; discuss temporary protection/analgesia

Document neurovascular state before/after splinting

Ischemia, disproportionate pain/progressive deficit: escalate

NICE NG39 major trauma

Chinese expert consensus on prehospital major trauma (2026)

Lower limb: local and systemic risk

Tiered recall; resident key-step simulation requires supervision and local policy.

Bleeding, open injury, neurovascular state and pain

Older low-energy hip injury: anticoagulation/baseline review

Recheck after splint/movement; verify destination/owner

Ischemia/deterioration: escalate; see domestic hip pathway

NHC geriatric hip fracture care and management guideline (2022)

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Bvm: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Two-person coordination, seal and chest response

Review oxygenation/ventilation and next airway plan

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Intubation: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Preparation, preoxygenation, backup plan and roles

Confirm placement/monitor; failed attempt needs rescue plan

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Cricothyrotomy: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Recognize failed intubation/oxygenation and call support

Trainer-based anatomy, equipment and effect checks

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Needle Decompression: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Integrate suspicion with respiratory/perfusion changes

Check effect; failure needs reassessment/escalation

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Tube Thoracostomy: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Indication, preparation, supervised key steps

Effect, drainage system, complications and review

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Efast: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Review standard windows; record quality/missing views

Negative scan does not exclude injury; use stability/repeat review

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Tourniquet: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Recognize life-threatening external limb bleeding

Control effect, time, distal state and handover

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Pelvic Binder: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Suspected pelvic injury/perfusion risk; local indications

Simulated placement, skin risks, effect and serial review

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma

Splinting: skill pocket

Supervised trainer practice; real procedures follow authorization and skill sources.

Document neurovascular checks before/after splinting

Escalate pain/perfusion changes or new sensory deficit

State authorization, supervision and rescue plan

OSCE: 0 absent / 1 prompted / 2 complete; flag critical errors

Chinese expert consensus on prehospital major trauma (2026)

NICE NG39 major trauma